Lab 6: Code and Branch Coverage Analysis

Scenario: Income Tax Calculator (income_tax.py) | **Course:** Software Engineering

1. Executive Summary & Coverage Concepts

Line Coverage measures whether each statement in the module was executed at least once during test execution. **Branch Coverage** is a stricter metric that evaluates whether every control flow decision point (e.g., if, elif, else) evaluates to both *True* and *False* across all runs. High line coverage can mask untested execution paths; branch coverage guarantees thorough decision path testing.

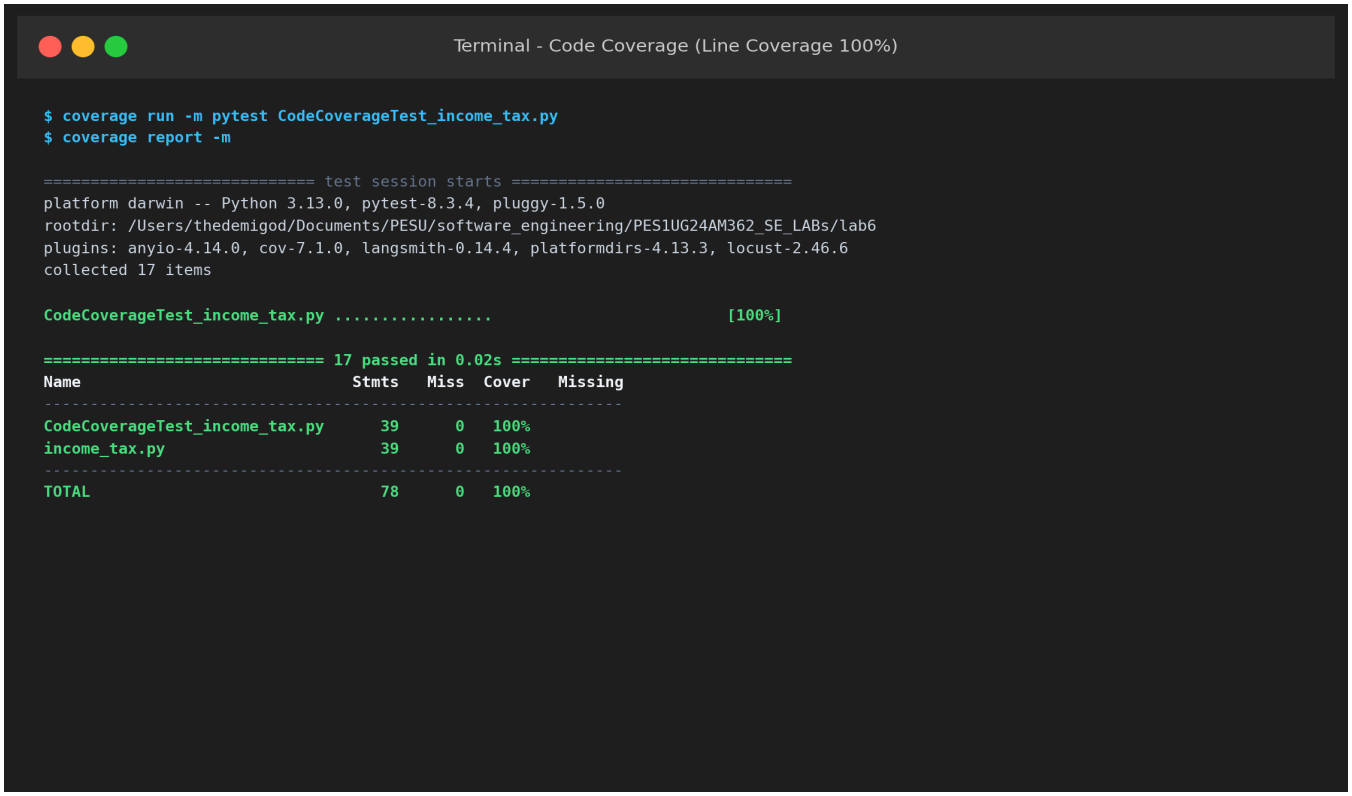
Metric	Initial Starter Test	Target	Achieved Result
Line (Code) Coverage	38% (24 missing statements)	>= 90%	100% (39/39 statements)
Branch Coverage	33% (7 missing partial branches)	>= 90%	100% (28/28 branches)
Test Suite Size	2 test cases	Comprehensive	27 test cases (All Passing)

2. Decision Branch Breakdown in income_tax.py

Function	Decision Point	Branch Conditions Tested	Resolution / Boundary
taxable_income	gross < 0	True -> ValueError('invalid income'); False -> proceeds	gross = -1 and gross = 0
taxable_income	regime check	'new' -> 75k ded; 'old' -> 50k+inv; else -> ValueError	Tested 'new', 'old', 'flat_tax'
taxable_income	investments cap	investments > 150k -> capped; <= 150k -> kept	Tested 250k, 150k, 80k
taxable_income	taxable < 0	True -> returns 0; False -> returns taxable	gross < deduction vs gross > deduction
tax_due	taxable < 0	True -> ValueError('invalid amount'); False -> proceeds	taxable = -1
tax_due	age categories	age >= 80 (500k exempt); age >= 60 (300k); else (250k)	Boundary ages: 85, 80, 70, 60, 59, 25
tax_due	taxable <= exempt	True -> returns 0; False -> calculates slab tax	taxable < exempt, taxable == exempt
tax_due	slab <= 250k	True -> slab * 5%; False -> 12500 + excess * 20%	slab = 200k, slab = 250k, slab = 500k
refund_status	difference > 0	True: on_time -> 'refund'; late -> 'refund delayed'	tax_paid > tax_due with True/False
refund_status	difference < 0	True: on_time -> 'pay balance'; late -> '...penalty'	tax_paid < tax_due with True/False
refund_status	difference == 0	Neither > 0 nor < 0 -> returns 'settled'	tax_paid == tax_due with True/False

3. Terminal Execution & Coverage Screenshots

Artifact 1: Code (Line) Coverage Analysis — 100% Achieved



Artifact 2: Branch Coverage Analysis — 100% Achieved (28/28 Branches)

